

Lab Exercise: Schematics with ESP32 - Part A Calculations

Circuit (a): 125 V supply with 20, 30 and 50 ohm resistors in series

1. Total resistance:

$$R_T = 20 + 30 + 50 = 100 \text{ ohms}$$

2. Total current:

$$I_T = V / R_T = 125 / 100 = 1.25 \text{ A}$$

3. Current through each resistor:

In a series circuit the current is the same everywhere, so $I = 1.25 \text{ A}$ through R1, R2 and R3.

4. Voltage drop:

$$V1 = 1.25 \times 20 = 25 \text{ V}$$

$$V2 = 1.25 \times 30 = 37.5 \text{ V}$$

$$V3 = 1.25 \times 50 = 62.5 \text{ V}$$

5. Power:

$$P1 = 25 \times 1.25 = 31.25 \text{ W}$$

$$P2 = 37.5 \times 1.25 = 46.875 \text{ W}$$

$$P3 = 62.5 \times 1.25 = 78.125 \text{ W}$$

$$\text{Total power} = 156.25 \text{ W} (= 125 \text{ V} \times 1.25 \text{ A})$$

Resistor	R (ohm)	I (A)	V (V)	P (W)
R1	20	1.25	25.0	31.25
R2	30	1.25	37.5	46.875
R3	50	1.25	62.5	78.125
Total	100	1.25	125.0	156.25

Circuit (b): 125 V supply with 20, 100 and 50 ohm resistors in parallel

1. Total resistance:

$$1/R_T = 1/20 + 1/100 + 1/50 = 0.05 + 0.01 + 0.02 = 0.08$$

$$R_T = 1 / 0.08 = \mathbf{12.5 \text{ ohms}}$$

2. Total current:

$$I_T = V / R_T = 125 / 12.5 = \mathbf{10 \text{ A}}$$

3. Current through each resistor:

$$I = V / R \quad I_1 = 125 / 20 = 6.25 \text{ A}$$

$$I_2 = 125 / 100 = 1.25 \text{ A}$$

$$I_3 = 125 / 50 = 2.5 \text{ A}$$

4. Voltage drop:

In a parallel circuit the voltage across each branch equals the supply, so $V_1 = V_2 = V_3 = \mathbf{125 \text{ V}}$

5. Power:

$$P = V \times I$$

$$P_1 = 125 \times 6.25 = 781.25 \text{ W}$$

$$P_2 = 125 \times 1.25 = 156.25 \text{ W}$$

$$P_3 = 125 \times 2.5 = 312.5 \text{ W}$$

$$\text{Total power} = 1250 \text{ W} (= 125 \text{ V} \times 10 \text{ A})$$

Resistor	R (ohm)	I (A)	V (V)	P (W)
R1	20	6.25	125	781.25
R2	100	1.25	125	156.25
R3	50	2.50	125	312.50
Total	12.5	10.00	125	1250.00